Accept the mandate with a 20% maximum drawdown tolerance.

Invest the USD 10 million as **AGG 55% ($5.5m) • GLD 20% ($2.0m) • SPY 15% ($1.5m) • XLE 5% ($0.5m) • XLK 5% ($0.5m)**, rebalanced quarterly: a 75% defensive core of bonds and gold, with 25% in equities for growth.

Why 20%: it is about five years of a 4% spending payout. Of 1,601 portfolios tested, none inside our AGG cap met a 15% tolerance, and the only ones that did held about 5% in equities and earned 5.2% a year, below today's 6.5% target.

We ask the committee to: approve the allocation and the 20% tolerance, and pre-approve lifting gold to 25% if the correlation trigger below fires and the new mix still passes every limit.

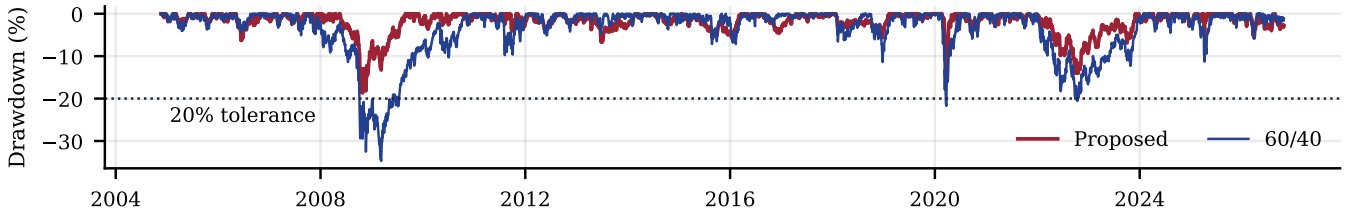
WHAT THE COMMITTEE GETS, AND EVERY LIMIT WE SET

	Proposed	60/40
Return (CAGR), 2004–2026	7.3%	8.1%
Return over T-bills	5.5%	6.4%
Volatility	7.3%	11.5%
Sortino ratio	1.07	0.83
Beta to US equities	0.27	0.60
Worst fall, 2004–2026	**−18.7%**	−34.7%
99% one-day ES (historical)	1.82%	3.12%
Chance of a >20% fall in 5 years	3.5%	33.0%

Check	Full sample	Limit	Headroom	Status
Excess return over T-bills	5.5%	≥ 2.5%	3.0 pp	Pass
Return against today's target	7.3%	≥ 6.5%	0.7 pp	Pass
Annualised volatility	7.3%	≤ 8%	0.7 pp	Pass
99% 1-day ES (conservative)	1.82%	≤ 2%	0.2 pp	Pass
Maximum drawdown, 2004–2026	18.7%	≤ 20%	1.3 pp	Pass
Deepest fall in a crisis replay	18.1%	≤ 25%	6.9 pp	Pass
Worst invented scenario	16.8%	≤ 25%	8.2 pp	Pass
99% VaR backtest (FHS), last 250 days	Green	Green	–	Pass
Allocation limits	Met	All met	–	Pass

ES (Expected Shortfall) is the average loss on the worst 1% of days; headroom uses unrounded figures. Limits are tested on the full sample; on 2020–26 data alone volatility (8.6%) and ES (2.17%) breach, driven by March 2020, and on 2021–26 data all four limits measurable on recent data pass.

The trade. The fund accepts about 0.9 percentage points a year less return than a 60/40 portfolio, and in exchange roughly halves the worst historical loss and cuts the average loss on the worst 1% of days by 42%. Among all mixes inside our limits that meet our return, the lowest achievable 99% ES is 1.76%; ours is 1.82%.



HOW IT BEHAVED WHEN MARKETS BROKE

	2008 crisis	2011 euro	2013 taper	2018 Q4	2020 COVID	2022 rates	Invented worst
Proposed	**−4.2%**	**−2.0%**	**−4.8%**	**−3.2%**	**−10.7%**	**−12.8%**	**−16.8%**
60/40 benchmark	−30.0%	−9.3%	−4.8%	−10.5%	−20.6%	−20.4%	−24.0%

Return over each crisis window, holding the starting weights. The deepest fall inside any window was −18.1%, in 2008 (60/40: −30.3%). The invented worst case is a deeper 2022 in which every holding, gold included, falls together. All are within the 25% stress limit.

THE COMMITTEE'S THREE QUESTIONS

What can go wrong? A shock in which bonds and shares fall together, so the defensive core stops defending. 2022 was the worst end to end replay (−12.8%); the invented breakdown costs −16.8%. Most of the tail risk sits in SPY (31.3%) and gold (27.9%), not in the bonds that hold most of the money.

How bad could losses become? On the worst 1% of days the average loss is $181,583 (1.82% of capital). The loss on a one in a thousand day is $248,510, and the average on days worse than that is $324,476. Across 3,000 simulated five year futures, the 1 in 20 worst fall is −18.7% and the chance of losing money is 1.4%.

How confident are we? The daily risk model was tested on 4,985 days it had never seen, from 2007: 59 breaches against 50 expected (Kupiec p = 0.21), never in the Basel red zone. Of nine methods compared, filtered historical simulation is the only one to pass the joint test of breach count and timing, marginally (p = 0.052).

RISKS TO THE RECOMMENDATION, AND WHAT WE WILL WATCH

Risks we are watching

- **The bond hedge broke in 2020.** A break test dates it to 26 March 2020; SPY–AGG correlation moved from −0.08 to 0.20, and is 0.43 in 2026.
- **Model errors cluster:** recent breaches bunched, so the 95% model is yellow today.
- **Return target:** a 17.7% chance of missing it over five years.

Pre-agreed triggers

- *Correlation:* if SPY–AGG stays above 0.20 for a quarter, move 5% from AGG to gold (gold to 25%) if that mix still passes every limit. **On current readings this comes to the first meeting.**
- *Risk budget:* if ES exceeds 2%, move equity into AGG in proportion to each fund's share of ES until it is back inside.
- *Model:* the 99% model yellow, or the 95% model yellow for two months, triggers a review.

Data: Yahoo Finance adjusted closes and FRED DTB3, 2004-11-19 to 2026-10-26 (5,517 trading days). Risk model: filtered historical simulation, tested out of sample from 2007-01-03. Methods and limitations: see the investment memo.